



Interpretable knowledge tracing via explicit–implicit alignment

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ARTICLE INFO

Keywords:

Interpretable knowledge tracing
Large language models
Student modeling
Personalized feedback

ABSTRACT

Interpretability in Knowledge Tracing (KT) is crucial for personalized learning. Current mainstream Deep Learning-based KT (DLKT) achieves strong predictive performance with ID-based representations, but students' latent knowledge states remain opaque to teachers and learners. Large Language Models (LLMs) can generate human-readable text, but explanations from existing LLM-augmented KT approaches are either detached from the predictive state or insensitive to individual trajectories. To bridge this gap, we propose **Interpretable Knowledge Tracing via Explicit–Implicit Alignment (IKT-EIA)**, which anchors implicit student knowledge representations with LLM-derived explicit textual insights. IKT-EIA consists of: (1) **Context-Guided Knowledge Integration (CGKI)** uses LLMs to generate trajectory-aware textual knowledge from student metadata; (2) **Query-Driven Knowledge State Alignment (QKSA)** treats the implicit state as a query to retrieve and fuse explicit semantics, producing an aligned knowledge state that supports subsequent explanation analysis; (3) **Personalized Explainability Adaptation (PEA)** generates structured and adaptive feedback by reading the aligned state together with raw interaction records, tailoring explanations to students' evolving mastery. Experiments on benchmark datasets show that IKT-EIA maintains competitive predictive performance while producing state-aware explanations that were found useful in a controlled evaluation setting.

1. Introduction

With the rapid growth of online learning platforms, KT has emerged as a fundamental task in intelligent education. KT enables personalized learning by modeling the evolution of students' knowledge states and predicting their future performance [1]. DLKT models, such as Deep Knowledge Tracing (DKT) [2], AKT [3], UKT [4], GIKT [5], and AMBER [6], employ architectures that include recurrent neural networks, attention mechanisms, and graph neural networks to achieve high prediction accuracy. These models encode problems, knowledge concepts (KCs), and student interactions as symbolic identifiers (IDs) and map them into a high-dimensional latent space, where learning dynamics can be implicitly captured. Despite their predictive success,

however, the black-box nature of these models limits interpretability [7]. Teachers and learners often struggle to understand how predictions are formed, making it difficult to provide targeted interventions or actionable feedback.

Existing approaches to Explainable KT (XKT), such as attention-based models and post-hoc interpretation techniques such as Shapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), have made initial progress toward enhancing transparency. However, these methods rely on abstract importance scores and coarse-grained visualizations, which provide limited actionable insights for educators and students and often fail to capture the causal mechanisms underlying student learning dynamics [3,8]. Most of them generate explanations that are presented as numerical

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<https://doi.org/10.1016/j.knosys.2026.116071>

Received 21 February 2026; Received in revised form 21 April 2026; Accepted 23 April 2026

Available online 27 April 2026

0950-7051/© 2026 Published by Elsevier B.V.

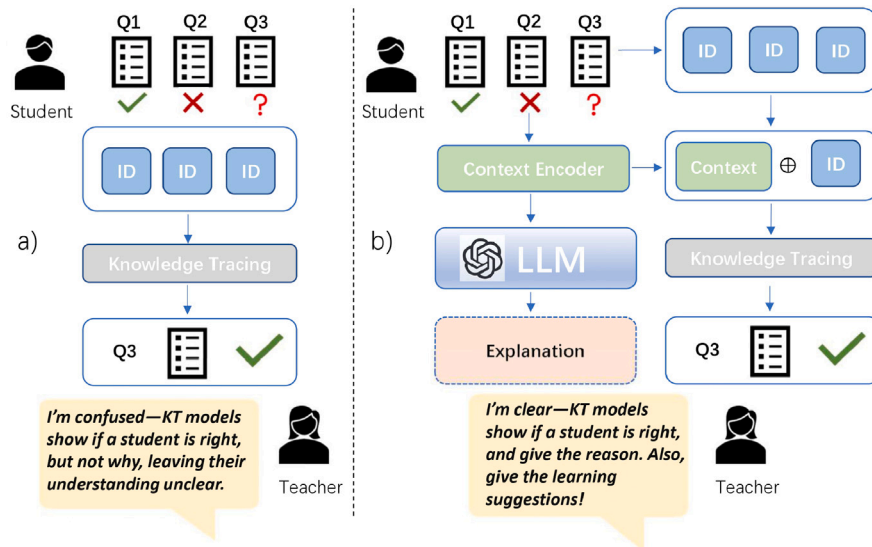


Fig. 1. (a) Traditional DLKT indicates correctness but lacks explanations, making understanding unclear for educators, who can see a student's performance but not the underlying reasons. (b) Enhancing KT with LLM-generated explanations provides actionable suggestions, transforming raw predictions into meaningful educational suggestions.

distributions and heatmaps rather than human-readable feedback. This makes it challenging for non-expert users to interpret the results or to translate them into effective instructional interventions. LLMs encode rich domain knowledge spanning education, cognitive science, and psychology, and can produce clear, context-sensitive natural-language explanations. By bridging the gap between model representations and human reasoning, LLMs offer the potential to make KT explanations not only more interpretable and readable but also actionable. This opens up new opportunities for building explainable KT systems (see Fig. 1).

Recent studies have explored LLM integration for KT explainability, building on these advantages. Most existing methods struggle to align explicit LLM-generated explanations with the implicit knowledge states modeled by DLKT, leading to two significant challenges. First, there is a disconnection between explanation generation and state prediction. Many approaches treat explanation as an isolated task, either by analyzing model representations post-hoc or by reasoning over student metadata, without aligning explanations with the internal states captured by DLKT [9,10]. Second, such misalignment leads to inconsistencies that undermine the model's interpretability and credibility. Current approaches rely on state-agnostic templates that cannot adapt to individual learning trajectories. They produce generic, one-size-fits-all explanations by overlooking students' evolving internal states. Interactive KT dialogue systems [11] provide broad feedback but lack adaptation to specific student contexts.

To address these challenges, we argue that future explainable KT models must satisfy two requirements. First, they should establish a tight coupling between prediction and explanation generation, ensuring that the explanations faithfully reflect the model's internal knowledge states. Second, they should exhibit personalized adaptability, enabling explanations to adjust to individual students' learning trajectories and evolving knowledge progression. Therefore, we propose the **Interpretable Knowledge Tracing via explicit-implicit Alignment (IKT-EIA)**. The core idea of IKT-EIA is to achieve explicit-implicit alignment, integrating natural language explanations provided by LLMs with the latent knowledge representations modeled by DLKT. IKT-EIA consists of three components:

1. **Context-Guided Knowledge Integration (CGKI):** Traditional KT models rely heavily on symbolic identifiers such as question and concept IDs, which lack semantic meaning and limit interpretability. CGKI addresses this issue by using LLMs to transform

students' interaction histories into contextualized textual descriptions that capture domain semantics. These descriptions are then encoded into semantic embeddings, producing human-interpretable knowledge representations that enrich the input space and lay the groundwork for pedagogically meaningful explanations.

2. **Query-Driven Knowledge State Alignment (QKSA):** Introducing semantic features alone may still lead to misalignment between explanations and the predictive process. The QKSA module employs a query-driven cross-attention mechanism in which the implicit ID-based state serves as the query and the semantically fused representations serve as keys and values. By letting the implicit state retrieve relevant explicit context from the interaction history, QKSA produces an aligned knowledge state that captures temporal dependencies and ensures consistency between predictions and explanations.
3. **Personalized Explainability Adaptation (PEA):** Building on the aligned explicit and implicit representations, the PEA module generates structured and personalized learning feedback. It categorizes students' knowledge into "strong areas," "areas for growth," and "areas needing improvement," while also incorporating historical patterns and predictive outcomes to produce fine-grained recommendations. Unlike generic template-based explanations, PEA emphasizes adaptability and actionability, offering tailored guidance designed to support both learners and instructors in making informed educational decisions.

In summary, our key contributions are as follows:

1. We propose IKT-EIA, a dual-model framework that connects DLKT prediction with LLM-based explanation through explicit-implicit alignment. IKT-EIA establishes a shared, aligned knowledge state, offering a consistent source for both prediction and explanation, and feeds the implicit state as a structured query into the explicit semantic space.
2. We design a knowledge-aligned framework: CGKI constructs trajectory-aware semantic context; QKSA performs query-driven alignment where the implicit state queries explicit semantics to form the aligned knowledge state; PEA generates structured, actionable feedback conditioned on this aligned state, improving controllability and faithfulness of explanations.

3. We validate IKT-EIA on multiple datasets to demonstrate that IKT-EIA achieves competitive predictive performance while enabling state-aware interpretability.

2. Related works

2.1. Deep learning-based knowledge tracing

Deep learning has advanced KT by capturing complex and non-linear patterns in sequential student data, surpassing traditional probabilistic approaches such as Bayesian Knowledge Tracing (BKT) [12] and Additive Factor Models (AFM) [13]. The seminal Deep Knowledge Tracing (DKT) model [2] introduced recurrent neural networks, specifically LSTMs, to model temporal dependencies in student responses, laying the foundation for modern DLKT research. Subsequent efforts extended DKT by incorporating KC representations [14], addressing the limitation of treating exercises as independent identifiers.

Memory-augmented KT. To capture long-term knowledge retention, memory-enhanced models such as DKVMN [7] introduced static and dynamic memory matrices to track and update students' mastery states. Later work explored adversarial learning [15] to improve model generalization and robustness by perturbing and regularizing latent states.

Attention-based KT. Attention mechanisms further advanced interpretability and performance by selectively focusing on past interactions most relevant to current predictions. SAKT [16] and AKT [3] employed self-attention to capture interaction relevance, forgetting, and question difficulty. SAINT [17] extended this idea with a transformer-based encoder-decoder architecture, enabling bidirectional modeling of student-question sequences. SimpleKT [18] simplified the attention formulation with dot-product attention while maintaining strong performance in time-aware learning scenarios.

Graph-based KT. Beyond sequence modeling, graph-based methods exploit relational structures among questions, KCs, and student interactions. GKT [19] and HGKT [20] modeled KC dependencies using graph structures, and GIKT [5] leveraged graph convolutional networks to refine question-KC correlations. PEBG [21] enriched question representations by pre-training question embeddings with side information from a question-skill bipartite graph. SGKT [22], DCKT [23], and TCKT [24] further improved KT by modeling exercise-skill structures, learner dynamics, and temporal-causal dependencies, respectively. KeenKT [25] addressed mastery-state ambiguity through distributional modeling for finer-grained estimation. Beyond KT, recent advances such as hierarchical graph perturbation (HiMod [26]) and adaptive modality interactions [27] suggest promising directions for sequence modeling.

2.2. Interpreting for knowledge tracing models

Interpretability is crucial for understanding how KT models generate predictions and for enabling their practical use in educational contexts. Existing efforts can be categorized into four lines of research.

Post-hoc local interpretability. These methods aim to explain individual predictions by analyzing the decision-making process after model inference. Techniques such as Layer-wise Relevance Propagation (LRP) [8,28] attribute prediction outcomes to specific input features, offering localized insights into student-KCs interactions. While effective for case-level analysis, these methods are retrospective and may not reflect the internal knowledge states that models have learned.

Global interpretability with explainable structures. Another line of work embeds meaningful cognitive theory into KT architectures. For example, EKT [29] incorporates multidimensional Item Response Theory (IRT) modules to capture latent student abilities, while neural temporal models [30,31] introduce interpretable temporal dynamics to model learning and forgetting processes. XKT [32] integrates cognitive

learning theories with a cognition processing network and an MIRT-based neural predictor, enabling concept-level knowledge tracing and more explainable predictions for questions involving multiple KCs. While aligning model architectures with educational and psychometric principles ensures the semantic traceability of latent representations, these structural priors limit the model's representational capacity to capture complex, nonlinear learning dynamics.

Global interpretability with explainable parameters. The third line focuses on transforming model parameters into interpretable cognitive constructs. Examples include IRT-like transformations [33], dynamic memory networks for tracking mastery [34], and recent probabilistic frameworks that link parameters to student ability estimation [35,36]. These approaches provide mappings between latent parameters and human-understandable concepts, but they typically capture static aspects of learning and may struggle to explain fine-grained dynamics.

LLM-enhanced interpretability. Recent studies have explored integrating LLMs to advance interpretability in KT [9–11,37–41]. By leveraging their broad knowledge of education, cognitive science, and psychology, LLMs can generate natural-language explanations that are more accessible to educators and learners. However, many existing LLM-augmented KT paradigms (e.g., LLM-KT [42]) treat LLM generation as either an independent post-processing step or a static auxiliary input, detached from the evolving latent states of the predictive model. This decoupling may lead to two issues: (1) static LLM features cannot track rapid knowledge transitions captured by the continuously updated DLKT state, and (2) without joint alignment, models may map response patterns to incorrect semantic anchors, producing unsupported or mastery-inconsistent feedback. The gap remains between human-readable explanations and the evolving implicit states learned by DLKT.

3. Methodology

As outlined above, existing KT models prioritize prediction over interpretability, creating a gap that limits their educational applicability. To bridge this gap, we design a framework that couples accurate prediction with personalized explanation. The methodology is organized into three parts, as is shown in Fig. 2. First, we introduce CGKI, which enriches raw interaction data with domain-specific semantics through LLM-based prompts. Building on this, QKSA aligns implicit latent states with explicit semantic embeddings via query-driven cross-attention, ensuring consistency between predictions and explanations. PEA then leverages the aligned states to produce trajectory-aware, structured feedback tailored to individual learners.

3.1. Problem statement

We begin with the problem statement, where we formalize the dual objectives of XKT as both response prediction and explanation generation.

Traditional KT models a student's evolving knowledge state, while XKT extends this by providing both prediction and interpretability. We use the Knowledge Concepts (KCs) throughout this paper to denote the knowledge skills or topic labels annotated in each dataset (e.g., "Solving Linear Equations" and "Graphing Negative Slopes"). Given a student S , their learning history is represented as an ordered sequence of T interactions: $S = \{s_j\}_{j=1}^T$, $s_j = \langle q_j, \{c_j\}, r_j, t_j \rangle$, where q_j denotes the question, $\{c_j\}$ represents the associated KCs, $r_j \in \{0, 1\}$ indicates response correctness, and t_j is the timestamp.

XKT optimizes (1) Prediction, estimating $P(\hat{r}^*|S, q^*)$, and (2) Explainability, generating $E^* = \text{LLM}(\text{prompt}(S, q^*))$, where prompt structures input for the LLM. XKT is formulated as $XKT = KT + \text{LLM-based explainability}$, integrating deep learning-based KT with LLMs to enhance prediction and interpretability for personalized learning.

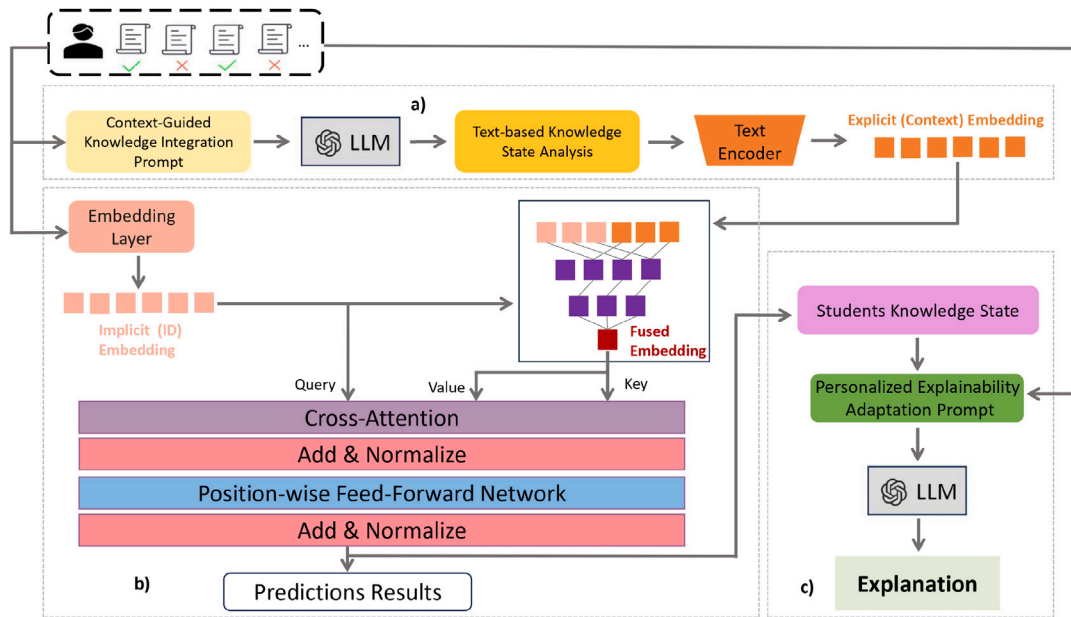


Fig. 2. Overview of IKT-EIA. The framework contains three key modules: (a) CGKI addresses the lack of human-interpretable input by generating contextualized textual knowledge from student metadata using LLMs (Section 3.2). (b) QKSA addresses the misalignment between ID-based implicit states and LLM-based explicit knowledge via query-driven attention and temporal alignment (Section 3.3). (c) PEA bridges aligned states with actionable, structured student feedback (Section 3.4).

3.2. Context-Guided Knowledge Integration (CGKI)

KT models rely on ID-based input features such as question and concept identifiers. While effective for capturing interaction patterns, these IDs carry no explicit meaning and do not reflect the conceptual relationships among educational items. This limitation not only limits the model's ability to reason about the domain structure but also renders its internal states opaque to both teachers and learners. The CGKI prompt transforms raw interactions into structured textual descriptions using an LLM, enriching the input with domain-aligned semantics and creating a foundation for more meaningful explanations. The prompt can be divided into three main components: (1) structuring student interaction history, (2) incorporating domain-specific contextual knowledge, and (3) generating standardized textual descriptions.

3.2.1. Structured student interaction history

We transform raw student interactions into a structured format, mapping each question ID to its corresponding KCs and response correctness (Fig. 3). Each interaction at time t is represented as $S_t = (QID_t, KC_t, Resp_t)$, where $Resp_t$ denotes correctness. The chronological history is structured as $H_n = \{S_1, S_2, \dots, S_n\}$.

3.2.2. Domain-specific contextual enhancement

Motivated by research [43–45], we consider a theory-informed partition of KC mastery into three levels: Weak, General, and Stable. Lower mastery values should reflect insufficient evidence of understanding, while higher values should indicate stable mastery.

In particular, we evaluate several moderate partitions around the theoretically motivated region and compare their predictive performance across datasets. The results show that the framework remains stable under moderate threshold variations, while the pair 0.25/0.75 provides the best overall balance across datasets. We adopt mastery scores at or below 0.25 as Weak, those at or above 0.75 as Stable, and the remaining ones as General.

In practical deployments, these thresholds can still be adjusted to align with local instructional standards.

3.2.3. Standardized textual representation

To make the generated descriptions comparable across interactions, we enforce a structured format for the LLM output that focuses on KC classification, excluding irrelevant information such as question IDs and other metadata. The standardized format is as follows: Stable Knowledge Concepts: {...}; General Knowledge Concepts: {...}; Weak Knowledge Concepts: {...}. Fig. 3 illustrates the complete version of the prompt.

3.2.4. Explicit encoding and fusion

Traditional ID-based representations capture only question-level interactions and lack explicit semantic relationships between knowledge states. To better capture semantics beyond IDs, we employ a semantic encoding approach that transforms LLM-generated text into high-dimensional embeddings using a frozen BERT [46] model to extract contextual features. These embeddings are then projected onto the hidden dimension of the knowledge tracing model. To unify implicit structural states with explicit knowledge, we fuse ID-based representations with explicit embeddings to ensure compatibility for downstream processing.

3.3. Query-Driven Knowledge State Alignment (QKSA)

Although CGKI introduces explicit knowledge, these representations are externally generated and remain decoupled from the KT model's internal latent states. Without a mechanism to unify them, explanations can become disconnected from actual predictions, leading to misleading feedback. QKSA addresses this by aligning two key representations: (1) the implicit latent states learned from ID-based sequences; (2) the explicit textual knowledge derived from CGKI via LLM prompts. QKSA employs a query-driven cross-attention mechanism: the implicit ID-based embedding acts as the query, while the fused embeddings that combine ID and semantic information serve as keys and values. Through this process, QKSA retrieves the most relevant grounded historical information and produces an aligned knowledge state, which is then shared by both response prediction and explanation generation.

[Context-Guided Knowledge Integration]
Below is the student's interaction history, containing question sequences and response patterns:
[Structured interaction history]
Question <question ID> involves knowledge concepts: <Knowledge concepts description list>
Student answered: <Correct/Incorrect>
[Student guessing effects]
In knowledge tracing tasks, students may sometimes guess when answering questions. In such cases, the student's answers are not based on true mastery of the knowledge but on random guessing. This can lead to unstable performance, where the student sometimes answers the same knowledge concepts correctly and other times incorrectly, which fails to accurately reflect the student's real knowledge level.
[Definition of Knowledge Concepts]
Additionally, knowledge tracing tracks a student's mastery over specific knowledge components (KCs). A knowledge component (KC) refers to a distinct skill or concept that the student needs to understand and master.
[Mastery Classifications]
Students' knowledge mastery can be divided into three states:
1. Stable knowledge concepts that the student has mastered, indicated by a correct response rate of over 75%;
2. Weak knowledge concepts, indicated by a correct response rate of below 25%;
3. General knowledge concepts, indicated by a correct response rate between 25% and 75%.
[Task Description]
Please generate a text description about the student's current knowledge mastery based on their answers. You need to categorize the knowledge concepts as stable, general, or weak according to the student's interaction history. Please strictly follow the format below and only provide the knowledge concepts descriptions, without extra explanations or background information.
[Output Structure]
Stable knowledge concepts: <List of stable knowledge concepts, separated by commas; if none, leave blank>. General knowledge concepts: <List of general knowledge concepts, separated by commas; if none, leave blank>. Weak knowledge concepts: <List of weak knowledge concepts, separated by commas; if none, leave blank>.
[Note]
Note: Do not add question IDs, explanatory text, or background information, only list the raw descriptions of the knowledge concepts.

Fig. 3. The detailed structure of the prompt for CGKI. The prompt is designed to guide the LLM through the semantic distillation step, transforming raw interaction logs into a structured, context-aware textual analysis by incorporating domain knowledge, such as student guessing effects and mastery classifications.

3.3.1. Query-driven cross-attention for knowledge state alignment

To align implicit and explicit knowledge along the temporal sequence, QKSA employs a cross-attention mechanism. At each time step t , the ID-based interaction embedding $\mathbf{i}_t = \text{Embed}(q_t, r_t)$ captures the student's implicit response pattern, while the fused embedding $\mathbf{z}_t = \mathbf{i}_t + \mathbf{e}_t^{\text{exp}}$ integrates the student-level semantic context produced by CGKI. The query is derived from the implicit representation, and the keys and values are derived from the fused representations:

$$\tilde{\mathbf{q}}_t = W_q \mathbf{i}_t, \quad \mathbf{k}_t = W_k \mathbf{z}_t, \quad \mathbf{v}_t = W_v \mathbf{z}_t, \quad (1)$$

where $W_q, W_k, W_v \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ are learnable projection matrices. As the query originates from $\mathbf{i}_t$, while the keys and values carry the fused semantic signal, the attention mechanism enables the implicit prediction pathway to actively retrieve semantically enriched historical context. Causal attention weights are computed as:

$$\alpha_{t,\tau} = \text{softmax}\left(\frac{\tilde{\mathbf{q}}_t^\top \mathbf{k}_\tau}{\sqrt{d_{\text{model}}}}\right), \quad \tau < t, \quad (2)$$

and the aligned knowledge state is obtained by aggregating the attended values:

$$\mathbf{h}_t = \sum_{\tau=1}^{t-1} \alpha_{t,\tau} \mathbf{v}_\tau. \quad (3)$$

The resulting $\mathbf{h}_t$ encodes both the student's temporal interaction pattern and the explicit semantic context, serving as the aligned knowledge state for both prediction and explanation.

3.3.2. Predicting next-step knowledge state

The aligned knowledge state $\mathbf{h}_t$ is used to predict the probability of a correct response for the next question:

$$\hat{y}_{t+1} = \sigma(W_p [\mathbf{h}_t; \mathbf{q}_{t+1}^{\text{emb}}] + b_p), \quad (4)$$

where $\mathbf{q}_{t+1}^{\text{emb}}$ denotes the embedding of the next question and $W_p \in \mathbb{R}^{1 \times 2d_{\text{model}}}$.

A key design is that both prediction and explanation are conditioned on the same aligned knowledge state $\mathbf{h}_t$ produced by QKSA. Since the explicit embeddings are precomputed by a frozen encoder and injected only through the key-value pathway, the projection matrices in QKSA can be learned end-to-end with the prediction objective without introducing an additional alignment loss.

3.4. Personalized Explainability Adaptation (PEA)

While previous components align implicit and explicit knowledge representations, they do not directly provide student-facing, interpretable feedback. PEA leverages the aligned knowledge states produced by QKSA to generate structured, personalized explanations. It uses LLMs as guided reasoners, operating on a tailored prompt that includes both the learning history and inferred mastery states. This design supports adaptive feedback aligned with each student's progression.

3.4.1. Personalized explanation prompt construction

We design a structured prompt template that integrates key student metadata and domain knowledge to guide explanation generation. The template consists of:

[Personalized Explainability Adaptation Prompt]
[Role Definition]
 As an educational analysis expert, please evaluate the student's knowledge mastery based on the following standardized profile data. Ensure your assessment considers both the student's profile and the 'Knowledge States' to evaluate the student's proficiency.
[Analysis Focus]
 Analysis Focus: Looking at: 1. How well you're doing with each topic. 2. Where you're making progress. 3. Which areas need more practice.
[Student Profile Construction]
 ID: {student_id} Your **Learning Record: Concepts** You've Studied: {kc_text_sequence} Your **Answers:** {responses} (1 = correct, 0 = incorrect) Your **Knowledge States:** {knowledge_states}
[Report Format]
 Your Learning Status: Your **Strong Areas** (Excellent Understanding): Topics you've mastered. Examples of what you're doing well. Ways to challenge yourself further. How you might help classmates. **Areas Where You're Growing** (Making Progress): Topics you're getting better at. What you're doing right. Specific parts to practice more. Tips to improve your understanding. **Areas to Focus On** (Need More Practice): Topics that need extra attention. Specific skills to work on. Clear steps for improvement. Study strategies that might help.

Fig. 4. The detailed structure of the prompt for PEA. This prompt is designed to be trajectory-aware, instructing the LLM to reason over the student's complete learning history (Your Learning Record) and inferred knowledge states (Your Knowledge States) to generate feedback on patterns of progress and areas for growth.

- **Student Profile:** Contains the student ID, learning history, responses, and knowledge states.
- **Learning Record:** Tracks the sequence of knowledge concepts studied, the correctness of responses, and inferred students' mastery description.
- **Standardized Output Format:** Enforces consistency in feedback, dividing it into Strong Areas, Areas for Growth, and Areas Needing More Practice.

Given a student's learning history H_n , where:

$$H_n = \{(KC_1, Resp_1), (KC_2, Resp_2), \dots, (KC_n, Resp_n)\}, \quad (5)$$

where KC_i represents a knowledge concept and $Resp_i \in \{0, 1\}$ indicates correctness, we then structure the input prompt as:

$$P = \{StudentProfile, H_n, \{Mastery(KC_i)\}_{i=1}^n, OutputFormat\}, \quad (6)$$

3.4.2. Adaptive explanation generation via LLMs

Using the structured prompt P , an LLM generates personalized explanations by reasoning over the student's learning history. Following the same mastery classification principle used in CGKI, we first compute a scalar mastery score for each KC. Let $\hat{p}_{c_i} \in [0, 1]$ denote the accumulated correctness rate of student responses on KC c_i , i.e., the fraction of correct answers among all interactions involving c_i . The mastery level is then determined by:

$$Mastery(c_i) = \begin{cases} \text{Strong,} & \text{if } \hat{p}_{c_i} \geq \theta_H \\ \text{Growing,} & \text{if } \theta_L < \hat{p}_{c_i} < \theta_H \\ \text{Needs Practice,} & \text{if } \hat{p}_{c_i} \leq \theta_L \end{cases} \quad (7)$$

where $\theta_H = 0.75$ and $\theta_L = 0.25$ are mastery thresholds following the same rationale described in the previous section.

The final explanation E_n is generated as:

$$E_n = \text{LLM}(P), \quad (8)$$

where the LLM adapts explanations based on mastery classification, ensuring personalized and interpretable feedback.

3.4.3. Structured output for interpretability

To maintain clarity, explanations follow a standardized format:

Strong Areas: {...}; Areas for Growth: {...}; Areas Needing More Practice: {...}.

This structure makes the explanations easier to understand across all students and provides them with clear, actionable insights into their strengths and areas for improvement. The complete prompt used for this process is shown in Fig. 4.

4. Experiments

In this section, we evaluate IKT-EIA through extensive experiments to address the following Research Questions (RQs):

- **RQ1:** Does IKT-EIA achieve competitive predictive performance compared with state-of-the-art DLKT models?
- **RQ2:** Is explicit-implicit alignment necessary, and does it contribute to both prediction accuracy and explanation faithfulness?
- **RQ3:** How sensitive is IKT-EIA to the choice of mastery thresholds, LLMs, encoders, and prompt design?
- **RQ4:** Are the generated explanations faithful, personalized, and useful?

4.1. Datasets and evaluation metrics

We conduct experiments on three datasets, **Algebra2005 (AL2005)**,¹ **ASSISTments2017 (AS2017)**,² and **Bridge2Algebra2006 (BD2006)**.³ These datasets, sourced from the KDD Cup 2010 and 2017 EDM Challenges, offer a diverse range of interaction scales and distributions of knowledge concepts, making them well-suited for assessing KT models under varying conditions. The details are as follows:

- **AL2005:** This dataset features step-level student interactions with mathematical problems, where each question is uniquely identified by concatenating the problem and step names. It consists of 884,098 interactions across 173,113 questions and 112 knowledge concepts (KCs), with an average of 1.3521 KCs per question. Additionally, it contains 4712 student sequences.
- **AS2017:** Comprising 942,816 interactions over 102 KCs, AS2017 is relatively compact in terms of question variety. This makes it suitable for evaluating medium-scale models while maintaining real-world applicability in educational settings.
- **BD2006:** Following a similar step-level question construction approach as AL2005, BD2006 provides a larger dataset with 1,824,310 interactions spanning 129,263 questions and 493 KCs, averaging 1.0136 KCs per question. It includes 9680 student sequences, making it an essential benchmark for large-scale knowledge tracing evaluations.

¹ <https://pslcdatashop.web.cmu.edu/KDDCup/>.

² <https://sites.google.com/view/assistmentsdatamining/dataset?authuser=0>.

³ <https://pslcdatashop.web.cmu.edu/KDDCup/>.

To ensure fair comparisons, we adopt standardized data preprocessing and splitting procedures, consistent with those used in prior DLKT models. Model performance is assessed using the area under the curve (AUC) and accuracy (ACC).

4.2. Baselines

To assess the effectiveness of IKT-EIA, we compared it against 13 state-of-the-art DLKT models and one LLM baseline (GPT-4o):

- **GPT-4o⁴**: Building on the evaluation methods of recent LLM-based approaches like LLM-KT, we organize the student's historical question-answer records and concept metadata into a structured natural-language instruction. GPT-4o uses its world knowledge and reasoning ability to predict the next response in a zero-shot setting.
- **DKT [2]**: An RNN (LSTM)-based model that predicts student performance by capturing historical interactions. It was among the first to demonstrate that deep sequential models outperform traditional probabilistic KT models.
- **DKT+ [47]**: An extension of DKT that introduces prediction-consistent regularization to address issues of overfitting and inconsistency; it encourages the model's outputs to be stable when student history is perturbed, improving robustness and generalization.
- **FoLiBiKT [48]**: Uses a linear bias/forgetting-aware component together with the dot-product between student ability and KC embeddings. It explicitly models forgetting and question/KC difficulty in a simpler, more parameter-efficient way.
- **Deep-IRT [34]**: Combines ideas from Item Response Theory (IRT) with deep neural architectures; it models latent student ability and item difficulty in a continuous space. This improves interpretability relative to pure black-box DLKT methods.
- **IEKT [49]**: Individual Estimation Knowledge Tracing adds two additional modules compared to standard RNN-based KT: a Cognition Estimation (CE) module before response prediction, which models how well a student cognitively understands a question, and a Knowledge Acquisition Sensitivity Estimation (KASE) module in the update ("write") stage, which models how much learning the student obtains from answering that question. IEKT achieves strong performance by accounting for student-question heterogeneity in both prediction and update phases.
- **DKVMN [7]**: Dynamic Key-Value Memory Networks that maintain a memory matrix of KCs and use value updates to track students' mastery over time, enabling the model to capture both retention and transitions among multiple KCs per interaction.
- **ATKT [15]**: An attention-based KT model enhanced through adversarial training. It seeks to improve generalization by introducing perturbations/regularization into the attention mechanism, which is useful in data-scarce and noisy environments.
- **GKT [19]**: Graph Knowledge Tracing that models relations between KCs via a graph structure, allowing relational influence between concepts/questions to affect predictions; this helps capture dependencies among skills.
- **SAKT [16]**: Self-Attentive Knowledge Tracing uses self-attention over past interactions to weight prior items by their relevance to the current question, helping model forgetting and context relevance.
- **SAINT [17]**: Builds on the transformer encoder-decoder architecture, treating student interaction histories as input sequences and modeling question/response pairs in both directions, enabling richer temporal context and cross-question relations.

- **AKT [3]**: Context-aware attentive KT models account for past interactions and temporal decay ("forgetting") in their attention weights, as well as question difficulties.
- **SimpleKT [18]**: Designed as a strong but lightweight baseline. SimpleKT combines question-specific embeddings (to model variation across questions within the same KC) with simple dot-product attention over past interactions to capture temporal awareness. It avoids complex modules for forgetting and, in empirical results, ranks among the top-3 in AUC across many datasets, showing that simpler architectures can still be very competitive.
- **KeenKT [25]**: A recent KT model that addresses knowledge mastery-state ambiguity through distributional modeling. It enables mastery estimation despite behavioral fluctuations, providing a strong baseline for tracking complex and uncertain learning dynamics.

4.3. Implementation details

Following prior DLKT studies, we randomly select 80% of student sequences for training and reserve the remaining 20% for evaluation, averaging results across 5-fold cross-validation. The model is optimized with AdamW for up to 200 epochs, using a staged learning-rate schedule: the rate is fixed for the first 40 epochs, halved between epochs 40 and 50, and then annealed to a minimum factor of 0.01 via cosine decay. We search over learning rates $[1 \times 10^{-5}, 3 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}, 3 \times 10^{-4}, 5 \times 10^{-4}]$ and weight decay $\{0.01, 0.001, 0.0001\}$, with a batch size of 512, embedding dimension of 768, dropout of 0.1, and early stopping with a patience of 20 epochs.

We use DeepSeek-R1-7B⁵ to generate student-level textual descriptions of knowledge states. A frozen bert-base-uncased⁶ model encodes these descriptions into explicit semantic embeddings, which are computed once per student and cached for reuse across all epochs and inference runs. The embeddings are aligned with the implicit latent states via QKSA. At inference time, cached embeddings are reused, so the prediction path retains the same complexity as a standard KT model. We use GPT-4o to generate human-readable student reports at the end of a learning period.

Training is conducted in PyTorch on a single NVIDIA A100-40G GPU with mixed precision.⁷

To assess the usefulness of explanations beyond case studies, we conduct a human evaluation on AL2005. We randomly sample 100 students and use their full interaction histories as inputs to the trained framework for report generation. Ten data science master's students serve as annotators. For each sampled student, reports are rated by three annotators in an anonymized, randomly ordered setting. Annotators score four dimensions on a 0–5 scale with 0.5 increments: Clarity, Specificity (KC-level), Actionability, and Personalization, and we report their mean as Usefulness Avg. We also compute KC-consistency between the KCs from student metadata and the KC mentions extracted from the generated report, using a synonym-aware Jaccard overlap (higher is better). Limitations of this annotator demographic are discussed in Section 5.

We also include Independent Generation (IG) as an explanation-generation baseline in the human evaluation: IG uses the same interaction inputs and report template as the human evaluator, but does not condition on the student's knowledge state.

To validate the semantic-state alignment, we conduct ablations across all three datasets under the same training setup and report AUC/ACC using the same 5-fold cross-validation for the following variants: the full model, w/o QKSA, w/o CGKI, and w/o PEA. To

⁴ <https://developers.openai.com/api/docs/models/gpt-4o>.

⁵ <https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-7B>.

⁶ <https://huggingface.co/google-bert/bert-base-uncased>.

⁷ Implementation

Table 1

Overall AUC (mean $\pm$ std) and ACC (mean $\pm$ std) performance of IKT-EIA and all baselines. For GPT-4o (direct), it is used in an instruction-following context that offers a binary Yes/No decision for each target interaction. The final performance is reported as the average and standard deviation of five folds.

Method Group	AS2017		AL2005		BD2006	
	AUC	ACC	AUC	ACC	AUC	ACC
DKT	0.7176 $\pm$ 0.0112	0.6874 $\pm$ 0.0092	0.9190 $\pm$ 0.0058	0.8636 $\pm$ 0.0045	0.7977 $\pm$ 0.0095	0.8543 $\pm$ 0.0072
DKT+	0.7193 $\pm$ 0.0109	0.6880 $\pm$ 0.0090	0.9169 $\pm$ 0.0060	0.8626 $\pm$ 0.0046	0.8033 $\pm$ 0.0092	0.8494 $\pm$ 0.0074
DKVMN	0.7080 $\pm$ 0.0113	0.6826 $\pm$ 0.0093	0.8054 $\pm$ 0.0108	0.8027 $\pm$ 0.0082	0.7983 $\pm$ 0.0096	0.8545 $\pm$ 0.0071
GKT	0.7208 $\pm$ 0.0108	0.6876 $\pm$ 0.0091	0.8110 $\pm$ 0.0105	0.8088 $\pm$ 0.0080	0.8046 $\pm$ 0.0091	0.8555 $\pm$ 0.0067
SAKT	0.6511 $\pm$ 0.0138	0.6638 $\pm$ 0.0112	0.7880 $\pm$ 0.0115	0.7954 $\pm$ 0.0088	0.7740 $\pm$ 0.0112	0.8461 $\pm$ 0.0075
SAINT	0.6889 $\pm$ 0.0125	0.6473 $\pm$ 0.0102	0.7775 $\pm$ 0.0118	0.7791 $\pm$ 0.0092	0.7781 $\pm$ 0.0108	0.8411 $\pm$ 0.0078
ATKT	0.7325 $\pm$ 0.0098	0.6930 $\pm$ 0.0082	0.9210 $\pm$ 0.0047	0.8630 $\pm$ 0.0044	0.7889 $\pm$ 0.0102	0.8511 $\pm$ 0.0073
AKT	0.7652 $\pm$ 0.0082	0.7124 $\pm$ 0.0065	0.8306 $\pm$ 0.0102	0.8124 $\pm$ 0.0078	0.8172 $\pm$ 0.0081	0.8505 $\pm$ 0.0072
SimpleKT	0.7586 $\pm$ 0.0088	0.7079 $\pm$ 0.0070	0.9048 $\pm$ 0.0068	0.8510 $\pm$ 0.0052	0.8096 $\pm$ 0.0089	0.8339 $\pm$ 0.0079
Deep-IRT	0.7061 $\pm$ 0.0115	0.6815 $\pm$ 0.0095	0.9115 $\pm$ 0.0065	0.8584 $\pm$ 0.0049	0.7957 $\pm$ 0.0098	0.8476 $\pm$ 0.0074
IEKT	0.7603 $\pm$ 0.0087	0.7119 $\pm$ 0.0061	0.8416 $\pm$ 0.0098	0.8236 $\pm$ 0.0075	0.8081 $\pm$ 0.0089	0.8494 $\pm$ 0.0073
FoLiBiKT	0.7612 $\pm$ 0.0078	0.7101 $\pm$ 0.0068	0.9229 $\pm$ 0.0052	0.8673 $\pm$ 0.0042	0.8099 $\pm$ 0.0088	0.8463 $\pm$ 0.0075
KeenKT	0.7661 $\pm$ 0.0069	0.7143 $\pm$ 0.0048	0.9396 $\pm$ 0.0041	0.8817 $\pm$ 0.0036	0.8271 $\pm$ 0.0074	0.8642 $\pm$ 0.0048
GPT-4o (direct)	–	0.6898 $\pm$ 0.0002	–	0.7094 $\pm$ 0.0004	–	0.6512 $\pm$ 0.0001
IKT-EIA (Ours)	0.7672 $\pm$ 0.0079	0.7152 $\pm$ 0.0062	0.9275 $\pm$ 0.0048	0.8719 $\pm$ 0.0038	0.8204 $\pm$ 0.0117	0.8567 $\pm$ 0.0168

further isolate the effect of semantic-state, we also include Misaligned Fusion (RF), which randomly shuffles the explicit embeddings within each mini-batch before fusion while keeping the corresponding implicit embeddings unchanged.

4.4. Comparison performance (RQ1)

Table 1 presents a comprehensive comparison between IKT-EIA and 14 baseline models. The results show that KeenKT achieves the strongest predictive performance on AL2005 and BD2006, while IKT-EIA attains the best overall results on ASSISTments2017 and outperforms the remaining baselines across all three datasets.

AL2005 features dense interaction records and a complex knowledge hierarchy. On this dataset, the purely predictive model KeenKT achieves the highest performance with an AUC of 0.9396. It uses unconstrained continuous distributional modeling, which is highly effective at capturing ambiguous, non-linear mastery fluctuations. On the other hand, IKT-EIA introduces a structural information bottleneck. By forcing the unconstrained continuous states to align with explicit semantic anchors via QKSA, it limits the degrees of freedom in the latent space. This alignment prevents the model from freely fitting uninterpretable, non-semantic behavioral fluctuations that pure predictive models exploit to maximize accuracy, slightly reducing its raw predictive power. The same situation occurs on BD2006. As a larger dataset with a much wider, more complex knowledge space (featuring 493 KCs compared to 112 in AL2005), BD2006 is sparser and more vulnerable to noise. Its learning trajectories are longer and more diverse, making the ability to capture behavioral variance even more essential for maximizing pure accuracy (where KeenKT achieves an AUC of 0.8271).

Despite this trade-off, IKT-EIA achieves competitive results on both datasets (e.g., an AUC of 0.8204 on BD2006) and shows improvements over traditional attention-based models like AKT. Although memory-based methods try to model forgetting, they are limited by the semantic sparsity of ID-only features. CGKI enhances the input with domain-specific contextual semantics, and QKSA combines them with implicit states to track interdependent concepts.

In ASSISTments2017, the modeling challenge differs from that of AL2005 and BD2006. Since the dataset contains a compact concept space (only 102 unique concepts), the room for improvement brought by highly flexible uncertainty modeling becomes smaller. Strong attention-based methods already capture most of the predictive patterns, so performance differences among the top models are very small. IKT-EIA achieves the best AUC and ACC, slightly surpassing both KeenKT and AKT. This suggests that when the concept space is relatively compact, and many interaction patterns are repeatedly observed, the semantic grounding provided by CGKI and the alignment mechanism of QKSA can still offer advantages.

We further assess IKT-EIA's robustness with respect to the choice of semantic encoder (BERT, RoBERTa, ALBERT, DistilBERT) and prompt design. All encoder variants yield consistent gains over ID-only baselines, with RoBERTa achieving the best results. Prompt ablations confirm that knowledge concepts are the most critical semantic signal, and surface-level modifications (punctuation, question IDs) have a negligible impact.

4.5. Ablation studies (RQ2)

We evaluate the necessity of alignment from predictive accuracy and explanation faithfulness. For prediction, we report AUC/ACC under module ablation; for explanation, we report KC-Jaccard, a synonym-aware overlap between dataset-provided KCs and KC mentions in the generated report.

As shown in Table 2, IKT-EIA achieves the best AUC and ACC on all three datasets. Among the ablated variants, removing PEA causes the smallest drop, as PEA primarily affects report generation. Removing CGKI or QKSA produces larger decreases, confirming that both semantic grounding and explicit-implicit alignment contribute to predictive performance. We further include Misaligned Fusion (RF) as a negative control, which randomly shuffles explicit embeddings within each mini-batch while keeping implicit states intact. RF performs worst across all datasets, showing that the gain depends on preserving the correct semantic-state correspondence enforced by QKSA.

As shown in Table 3, the full IKT-EIA achieves the highest JC across all datasets, with scores of 0.438 on AS2017, 0.428 on AL2005, and 0.369 on BD2006. Removing PEA results in the smallest decrease, followed by CGKI, QKSA, and RF. On AL2005, JC decreases from 0.428 to 0.409 (w/o PEA), 0.371 (w/o CGKI), 0.319 (w/o QKSA), and 0.272 (RF). The ranking of ablation variants (IKT-EIA > w/o PEA > w/o CGKI > w/o QKSA > RF) is identical across all three datasets despite their different structural characteristics (112 KCs in AL2005, 102 KCs in AS2017, and 493 KCs in BD2006). The relative JC decrease caused by removing PEA remains stable, with scores approximately dropping from 0.438 to 0.421 on AS2017, from 0.428 to 0.409 on AL2005, and from 0.369 to 0.357 on BD2006. This suggests that the generalizability of PEA may stem from the broader advantage of conditioning report generation on aligned knowledge states.

To illustrate the effect of misalignment, we compare the generated reports for the same student under the two conditions. As shown in Table 4, evaluators, when presented with the student's known KC trajectory together with the generated reports, judged the RF report to deviate from the student's actual learning profile. While the aligned model produces feedback grounded in the student's observed KC history, the

Table 2

Module ablation across three datasets. RF stands for Misaligned Fusion. We report AUC and ACC (mean $\pm$ std).

Variant	AS2017		AL2005		BD2006	
	AUC	ACC	AUC	ACC	AUC	ACC
w/o PEA	0.7659 $\pm$ 0.0081	0.7140 $\pm$ 0.0064	0.9248 $\pm$ 0.0052	0.8690 $\pm$ 0.0042	0.8188 $\pm$ 0.0121	0.8550 $\pm$ 0.0169
w/o CGKI	0.7588 $\pm$ 0.0088	0.7077 $\pm$ 0.0070	0.9222 $\pm$ 0.0056	0.8659 $\pm$ 0.0046	0.8094 $\pm$ 0.0130	0.8348 $\pm$ 0.0178
w/o QKSA	0.7571 $\pm$ 0.0091	0.7061 $\pm$ 0.0072	0.9189 $\pm$ 0.0063	0.8628 $\pm$ 0.0051	0.8078 $\pm$ 0.0134	0.8316 $\pm$ 0.0183
RF	0.7546 $\pm$ 0.0096	0.7033 $\pm$ 0.0077	0.9149 $\pm$ 0.0069	0.8583 $\pm$ 0.0056	0.8049 $\pm$ 0.0141	0.8279 $\pm$ 0.0191
IKT-EIA	0.7669 $\pm$ 0.0080	0.7145 $\pm$ 0.0063	0.9272 $\pm$ 0.0049	0.8715 $\pm$ 0.0039	0.8201 $\pm$ 0.0118	0.8563 $\pm$ 0.0168

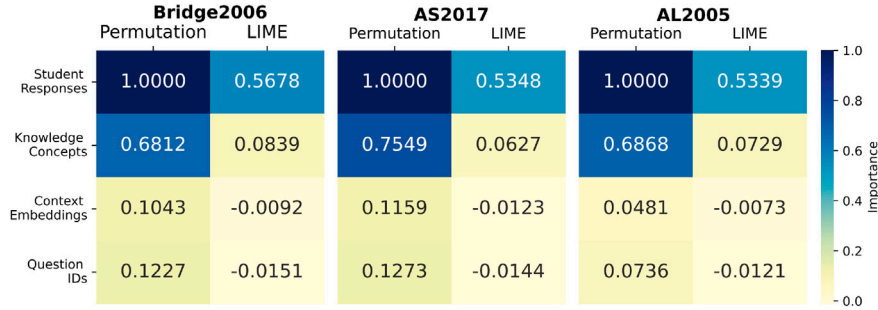
LIME and Permutation Analysis Comparison

Fig. 5. Feature importance analysis using LIME and Permutation methods. Student response patterns dominate predictive decisions, while semantic features serve as anchors that enable state-aware explanations.

Table 3

Module ablation results on JC (KC-Jaccard) across three datasets. JC measures explanation faithfulness by comparing the knowledge concepts contained in the generated report with dataset-provided concept annotations. Higher values are better.

Variant	AS2017	AL2005	BD2006
w/o PEA	0.421 $\pm$ 0.018	0.409 $\pm$ 0.037	0.357 $\pm$ 0.020
w/o CGKI	0.381 $\pm$ 0.021	0.371 $\pm$ 0.044	0.314 $\pm$ 0.023
w/o QKSA	0.339 $\pm$ 0.024	0.319 $\pm$ 0.017	0.279 $\pm$ 0.027
RF	0.292 $\pm$ 0.026	0.272 $\pm$ 0.031	0.244 $\pm$ 0.029
IKT-EIA	0.438 $\pm$ 0.016	0.428 $\pm$ 0.017	0.369 $\pm$ 0.018

RF report retains only partial overlap, weakens stronger competencies, and introduces unsupported weaknesses such as Word-Problem Translation and Mathematical Modeling.

This qualitative gap is reflected not only in RF's lower KC-Jaccard score but also in the generated explanation. When explicit students' knowledge states do not align with the student's implicit understanding, the report may look fluent, but no longer represents what the student actually mastered. At the same time, removing CGKI and PEA results in smaller decreases than removing QKSA or applying RF, indicating that these components primarily affect the richness and personalization of the generated report, whereas QKSA plays a more central role in preserving KC-level consistency. On AL2005, removing QKSA decreases AUC (from 0.9272 to 0.9189), but has a much larger effect on JC (from 0.428 to 0.319). This suggests that alignment contributes more to explanation faithfulness than to predictive performance (see Fig. 5).

We also apply LIME and permutation importance to understand which inputs most influence predictions. The results indicate that student responses consistently rank as the most critical feature, serving as the strongest predictive signal in the analysis. This aligns with the expectation that historical correctness patterns carry the highest information density for next-step prediction, and the ID-based representation in DLKT captures these regularities.

Although context embeddings receive lower attribution weights, this does not imply that semantics are unnecessary. In IKT-EIA, semantic information is introduced primarily to anchor the latent knowledge state with interpretable context, rather than to dominate predictive

accuracy. QKSA injects semantic anchors into the aligned state and is read by the explanation module, improving the quality of the generated feedback for each student. Without semantic anchoring, explanations would rely largely on IDs and response patterns, which tend to generate more generic, less concept-specific feedback. The attribution results support that IKT-EIA preserves strong KT predictive signals while enabling state-aware explanations.

4.6. Sensitivity analysis (RQ3)

We analyze sensitivity across four aspects, covering how different mastery thresholds, LLMs, encoders, and prompts affect IKT-EIA's performance.

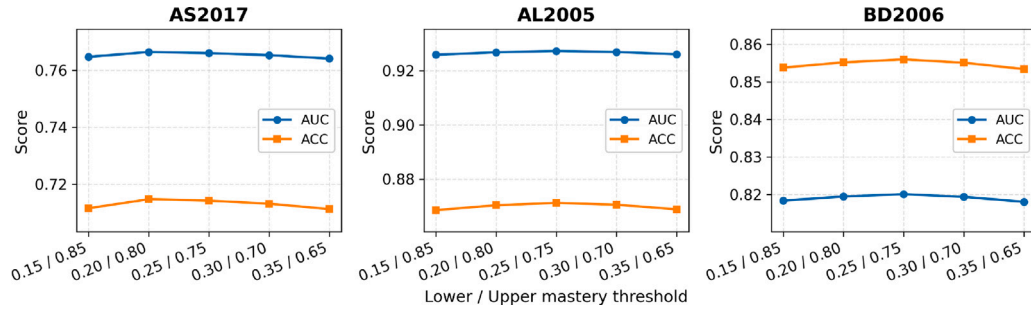
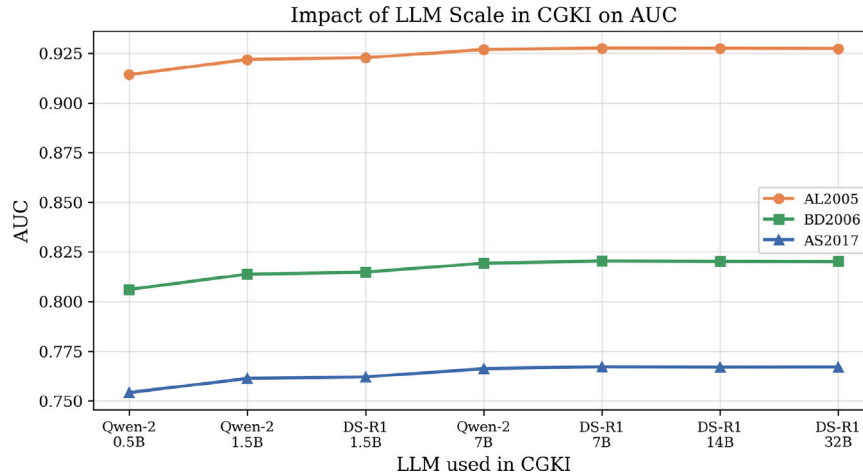
Impact of mastery thresholds. To justify the lower and upper mastery thresholds used in CGKI, we evaluate several candidate threshold pairs across the three datasets. As shown in Fig. 6, the overall performance remains stable within a moderate threshold range, while more extreme settings lead to slightly weaker results. On AL2005 and BD2006, 0.25/0.75 yields the best performance among the tested settings, whereas on AS2017, 0.20/0.80 is slightly better, and 0.25/0.75 remains very close. The differences among 0.20/0.80, 0.25/0.75, and 0.30/0.70 are small across all three datasets, indicating that the framework is not sensitive to moderate threshold changes. The minor dataset-dependent differences suggest that the best threshold pair may depend on task characteristics and data distributions, and these thresholds should be adjusted in practical applications based on specific environments. We adopt 0.25/0.75 as the default setting because it offers the best overall trade-off across datasets.

Impact of LLMs used in CGKI. We examine how the choice of LLM in CGKI affects performance on AS2017, AL2005, and BD2006. Fig. 7 shows that scaling the CGKI LLM from smaller to larger models improves both AUC and ACC across all datasets, with gains saturating at 7B. DeepSeek-R1-7B achieves near-best results on AL2005 and BD2006, while larger models (14B and 32B) show marginal improvements. This suggests that stronger LLMs can extract more informative semantics from interaction histories, and that a medium-sized LLM can already meet the semantic extraction requirement.

Table 4

Illustrative comparison of report excerpts for the same AL2005 student (Fig. 8 and Fig. 9). IKT-EIA identifies KC-level strengths and weaknesses that align with the student's history, whereas the misaligned setting (RF) exhibits partial overlap.

	IKT-EIA	RF
Strong Areas	Identifying Units, Defining Variables	Defining Variables
Areas for Growth	Graphing Concepts (improving trend noted)	Graph Feature Interpretation (downgraded from a stronger area)
Areas Needing Practice	Solving Equations with Negative Coefficients, Graphing Negative Slopes	Word-Problem Translation, Mathematical Modeling

**Fig. 6.** Sensitivity analysis of the lower/upper mastery thresholds across three datasets.**Fig. 7.** AUC as a function of LLM scale in CGKI. Performance improves rapidly up to 7B parameters and plateaus thereafter, indicating that a medium-sized LLM suffices for semantic extraction.

Semantic encoder compatibility. We replace the semantic encoder in IKT-EIA with four widely used pre-trained language models: BERT-base, RoBERTa, ALBERT-base-v2, and DistilBERT-base. Table 5 demonstrate three findings: (1) semantic enrichment leads to universal gains. Regardless of the encoder used, integrating textual semantics improves both AUC and ACC relative to ID-only baselines. This confirms that CGKI distills human-readable semantics into features that QKSA can align with latent states. (2) RoBERTa outperforms the other encoders, achieving the best results on AS2017 (AUC = 0.7708, ACC = 0.7189) and AL2005 (AUC = 0.9282, ACC = 0.8731). Its advantage lies in the richer contextual representations that RoBERTa produces, which are beneficial for datasets with dense conceptual hierarchies. (3) Lightweight encoders remain competitive. Both ALBERT and DistilBERT achieve strong results across all datasets, indicating that semantic augmentation is more critical than model size.

Prompting strategy ablation. We evaluate the contributions of the different components within the CGKI by removing or modifying their content. Only the CGKI prompt content is modified, and the underlying interaction records used by the predictor remain unchanged. The results in Table 6 show that the model is robust to prompt perturbations, but

remains sensitive to the removal of core semantic signals. Removing KCs results in the largest decline in performance, highlighting their central role in modeling students' mastery states. Removing student answers also reduces performance (AUC = 0.9266), indicating that students' responses provide useful behavioral evidence. Removing question IDs, background information, or modifying punctuation causes minor changes, suggesting that the model relies mainly on informative prompt content. Beyond predictive performance, Table 6 reveals how different CGKI prompt components affect the final reports. Since KC-Jaccard measures the overlap between the KCs reflected in the generated report and those supported by the student's learning history, it provides an indirect view of how well CGKI-derived semantic cues are retained through the downstream QKSA alignment and PEA generation stages. Removing KCs results in the largest drop in KC-Jaccard, from 0.431 to 0.386, indicating that KCs are the primary anchors in the final report. Removing student answers also reduces KC-Jaccard to 0.398, suggesting that although KC mentions remain available, the loss of response evidence weakens the model's ability to maintain mastery-consistent. Removing question IDs and background information, and modifying punctuation, leaves KC-Jaccard nearly unchanged, indicating that the final report depends mainly on semantic cues under the current CGKI ablation setting.

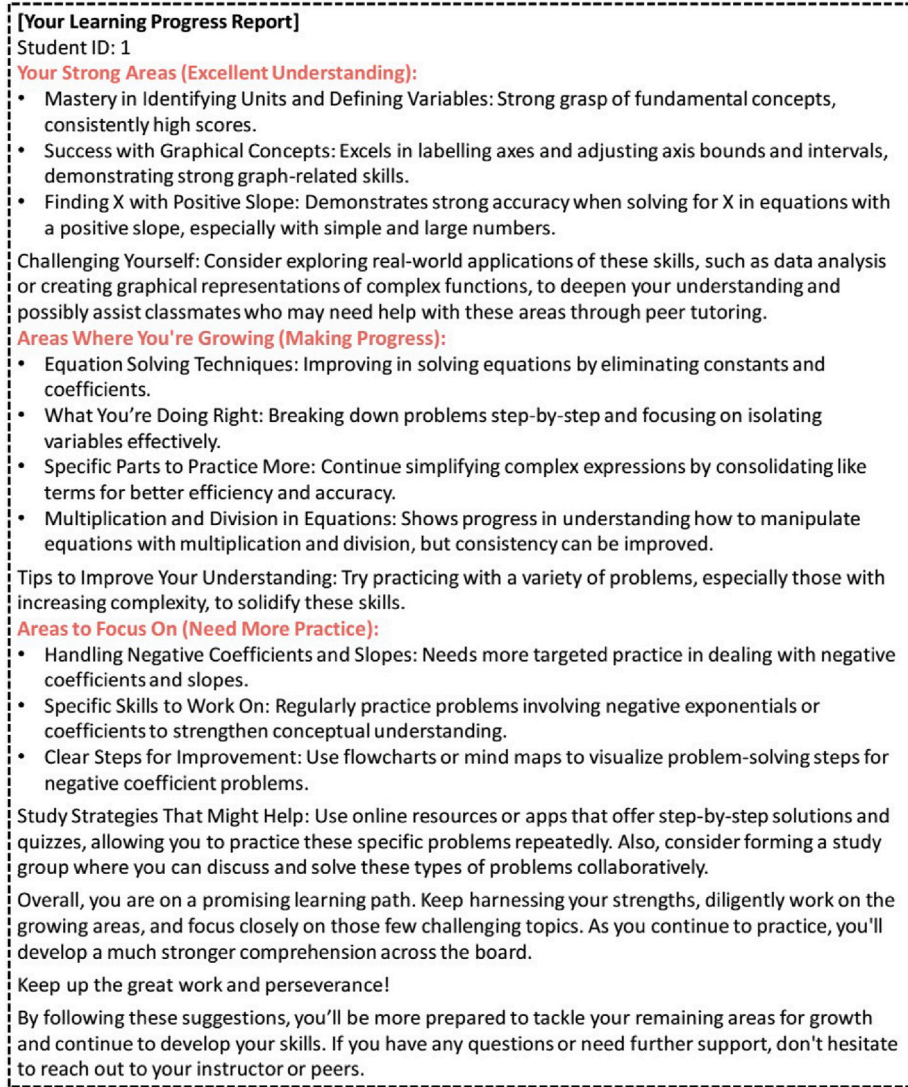


Fig. 8. A complete personalized learning report generated by PEA. It details the student's strengths and weaknesses and provides targeted practice tips and strategies to address specific learning gaps.

Table 5

Impact of Different Semantic Models on IKT-EIA Performance. The table shows the resulting AUC and ACC when using different pre-trained language models (BERT, RoBERTa, ALBERT, and DistilBERT) as the semantic encoder component across the three benchmark datasets.

Encoder	AS2017		AL2005		BD2006	
	AUC	ACC	AUC	ACC	AUC	ACC
BERT	0.7672	0.7152	0.9275	0.8719	0.8204	0.8567
RoBERTa	0.7708	0.7189	0.9282	0.8731	0.8241	0.8592
ALBERT	0.7635	0.7118	0.9261	0.8694	0.8172	0.8538
DistilBERT	0.7628	0.7106	0.9258	0.8688	0.8165	0.8531

Impact of LLMs in PEA. We further analyze the impact of using different LLMs as the report generator on AL2005. Table 7 shows a trend that stronger LLMs, especially closed-source LLMs (e.g., GPT-4o), lead to better explanation faithfulness and human-perceived quality. Larger LLMs achieve higher KC-Jaccard scores and higher ratings in Clarity, Specificity, Actionability, and Personalization. Among open-source LLMs, explanation quality improves with increasing model size, which aligns with the intuition that greater generation capacity helps the

model produce more specific feedback. A stronger generator can further amplify the benefits brought by semantic-state alignment.

4.7. Explanation quality (RQ4)

Having established explanation faithfulness via KC-Jaccard in Section 4.5, we then evaluate whether the generated reports are personalized and useful.

[Your Learning Progress Report]

Student ID: 1

Your Strong Areas (Excellent Understanding):

- Understanding Variable Setup: You seem to be comfortable identifying variables and recognizing what quantities need to be tracked in a problem. This suggests a reasonable grasp of how to begin structured mathematical tasks.
- How to Build on This Strength: Try applying this same clarity in setup when working on more complex multi-step problems or problems that combine symbolic and graphical information.

Areas Where You're Growing (Making Progress):

- Interpreting Graph Features: You appear to be making progress in reading graph features such as slope direction, axis labels, and interval changes. Continued practice should help make this understanding more stable.
- What You're Doing Right: You seem to benefit from tasks that allow you to compare visual patterns with mathematical expressions.
- Specific Parts to Practice More: Keep practicing how graph changes reflect equation changes, especially when moving between tables, equations, and plotted lines.
- Tips to Improve Your Understanding: Using graphing tools or sketching quick visual summaries may help strengthen these connections.

Areas to Focus On (Need More Practice):

- Word-Problem Translation and Mathematical Modeling: You may need more support in translating verbal descriptions into equations and selecting the right operations to represent relationships.
- Specific Skills to Work On: Practice identifying which information in a problem statement is essential and how it maps to algebraic expressions.
- Clear Steps for Improvement: Before solving, underline key quantities in the problem and decide what each one represents mathematically.
- Study Strategies That Might Help: Guided examples and short exercises on equation setup may help you become more confident in modelling word problems.

Overall, you are showing positive development in several core areas. Strengthening the connection between symbolic setup, visual interpretation, and problem translation will help you become a more flexible problem solver. Keep working steadily and stay confident in your progress! By continuing to practice these areas, you will be better prepared for more complex tasks in the future. If needed, seek additional help on translating problem statements into equations and interpreting graphical changes.

Fig. 9. A personalized learning report generated under RF. Compared with the aligned version, the report remains fluent but becomes less faithful to the student's actual KC trajectory, showing semantic drift and partially unsupported mastery judgments.

Table 6

Prompting strategy ablation on AL2005. We remove key elements to assess robustness in predictive AUC and ACC under the same training setup.

Prompt	AUC	ACC	KC-Jaccard
IKT-EIA w/o student answers	0.9266	0.8634	0.398
IKT-EIA w/o question ID	0.9271	0.8629	0.427
IKT-EIA w/o background information	0.9273	0.8646	0.429
IKT-EIA w/o knowledge concepts	0.9269	0.8607	0.386
IKT-EIA w/ punctuation modification	0.9272	0.8623	0.430
IKT-EIA	0.9275	0.8719	0.431

We conduct a human evaluation on AL2005 to validate the usefulness of the generated report. Table 8 shows that the IKT-EIA outperforms IG on all four metrics. The improvements are mainly in dimensions that require aligning the report with the student's learning state. In particular, Specificity increases from 3.22 to 4.02, and Actionability increases from 3.14 to 3.95, indicating that conditioning on the

student's knowledge state provides more specific guidance than generic feedback. The largest gains in Personalization also reflect the benefit of using the student's knowledge state. Despite this, the minimum difference is in Clarity, which also shows that both methods can produce readable reports under the same report template, and IKT-EIA improves content quality. These gains bring a higher Usefulness Avg (from 3.39 to 4.11). Human ratings indicate that explicit-implicit alignment can make report generation more personalized to each student's interaction history.

The case study shown in Fig. 8 highlights the strengths of the PEA component. It offers structured explanations that categorize a student's knowledge into strong areas, areas for growth, and focus areas, providing multi-level cognitive insights. Unlike post-hoc feature attribution methods (e.g., Perturbation and LIME), PEA adapts its feedback to the student's evolving learning history. For example, the complete learning progress report for a sampled student provides a detailed analysis of their strengths and areas for improvement.

Table 7
Impact of LLMs in PEA on AL2005.

LLMs	KC-Jaccard	Human evaluation			
		Clarity	Specificity	Actionability	Personalization
LLama-3-8B	0.408	3.72	3.55	3.18	3.46
Qwen-2-7B	0.412	3.86	3.41	3.28	3.39
GLM-4-9B	0.422	3.95	3.64	3.41	3.62
DeepSeek-R1-7B	0.424	4.01	3.72	3.48	3.58
GPT-4o	0.431	4.35	4.12	3.95	4.21

Table 8
Human evaluation on AL2005 (100 sampled students; 3 annotators per report). Scores are on 0–5 with 0.5 increments. IG stands for Independent Generation. We report mean $\pm$ std.

Method	Clarity	Specificity	Actionability	Personalization	Usefulness Avg
IG	4.15 $\pm$ 0.58	3.22 $\pm$ 0.75	3.14 $\pm$ 0.72	3.05 $\pm$ 0.78	3.39 $\pm$ 0.61
IKT-EIA	4.28 $\pm$ 0.52	4.02 $\pm$ 0.56	3.95 $\pm$ 0.60	4.18 $\pm$ 0.55	4.11 $\pm$ 0.49

The report highlights the student's proficiency in identifying units, defining variables, and graphing concepts, while also recommending additional practice in solving equations and addressing negative coefficients and slopes.

Instead of generic explanations, PEA offers targeted interventions that address individual learning gaps. The personalized feedback provides specific, actionable recommendations, such as additional practice with linear and quadratic equations, with a focus on negative coefficients.

PEA's feedback extends beyond simple knowledge classification to include instructional strategies. It offers targeted techniques to improve efficiency and accuracy, such as using "flowcharts or mind maps to visualize problem-solving steps" and adopting step-by-step problem-solving and visualization for graphing negative slopes. Unlike generic feedback, which may indicate only "needs improvement", this detailed report provides a clear roadmap tailored to the student's unique learning needs. This level of customization has the potential to enhance the learning experience by addressing students' strengths and weaknesses, thereby supporting more targeted learning.

5. Limitations

Despite the promising results, this study has two main limitations. First, IKT-EIA introduces additional computational cost at the preprocessing stage through LLM-based generation in CGKI. However, because the resulting semantic embeddings are computed once per student and cached for reuse across all training epochs and inference runs, the actual overhead during training and inference is slightly larger than that of a standard DLKT model. Computational efficiency was not a focus of this work, and a systematic runtime and memory benchmark is left for future study. Second, the human evaluation is conducted in a lab-based setting with data mining students on a single dataset (AL2005). Extending such evaluations to authentic classroom settings with educational practitioners remains another important future direction.

6. Conclusion

In this work, we revisit the gap between accurate knowledge tracing and actionable, personalized educational feedback, and propose IKT-EIA, a unified framework that bridges DLKT prediction and LLM-based explanation through explicit-implicit alignment. IKT-EIA builds a shared, aligned knowledge state with QKSA and reuses it to support next-step prediction and state-aware feedback generation. Extensive experiments on multiple benchmark datasets show that IKT-EIA maintains competitive predictive accuracy while producing coherent, learner-specific explanations that were rated as useful in a controlled evaluation setting. IKT-EIA provides a pathway for integrating predictive KT models with explanation-oriented LLMs within an aligned representation, and validating its utility in real-world classrooms remains an important direction for future work.

CRedit authorship contribution statement

Ersheng Ni: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jiaxin Wang:** Validation, Formal analysis. **Jining Luan:** Validation, Data curation. **Jiacheng Liu:** Validation, Data curation. **Shuyan Xiong:** Writing – review & editing, Validation. **Sibo Xu:** Investigation, Data curation. **Weihua Cheng:** Writing – review & editing, Resources. **Mengmeng Chen:** Validation, Resources. **Yongxin Ni:** Validation, Resources. **Youhua Li:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The work described in this paper was partially supported by the InnoHK initiative, the Government of the HKSAR, and the Laboratory for AI-Powered Financial Technologies.

Data availability

Data will be made available on request.

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